MODEL EVALUATION REPORT

Project Overview

The Telecom KPI Anomaly Detection Dashboard is an AI-powered application developed to automatically identify abnormal behavior in telecom network Key Performance Indicators (KPIs). The project applies to the Isolation Forest algorithm to detect anomalous KPI records without requiring labelled training data.

The detected anomalies are displayed through an Interactive Streamlit dashboard to assist network engineers in monitoring network health and identifying potential issues.

Dataset Description

The dataset contains telecom network KPI records collected from multiple network cells

Dataset Information

Parameter	Description
Dataset	Telecom KPI Dataset
Format	CSV
Records	1000
Target	Anomaly Detection
Learning Type	Unsupervised Learning

Features Used

The following features were used for anomaly detection.

- RSRP
- SINR
- Throughput
- Latency
- Packet Loss
- Connected Users
- Signal Quality Score
- Network Load
- Network Health Index

Data Preprocessing

The preprocessing stage included:

- Missing value handling
- Duplicate removal
- Invalid record removal
- Timestamp conversion
- Dataset validation

Feature Engineering

Three additional features were created

- Signal Quality Score

0.6*SINR+0.4*(RSRP+120)

- Network Load

Connected Users/200

- Network Health Index

Throughput – Latency - (10*Packet Loss)

Feature Scaling

Before training, all numerical features were standardized using StandardScaler.

Scaling ensures that each feature contributes equally to the Isolation Forest algorithm and prevents features with larger numerical ranges from dominating the model.

Model Information

Parameter	Value
Algorithm	Isolation Forest
Number of Trees	100
Contamination	0.05
Random State	42

Hyperparameter Tuning

The following parameter combinations were evaluated.

Number of Trees

- 50
- 100
- 200

Contamination Values

- 0.01
- 0.03
- 0.05
- 0.10

The final model used:

Trees=100

Contamination=0.05

Performance Metrics

Metric	Value
Accuracy	0.7720
Precision	0.4200
Recall	0.0955
F1 Score	0.1556

Model Output

The trained model produces:

- Normal records
- Anomalous records

The prediction results are stored in:

Data/processed/telecom_kpi_with_anomalies.csv

Each record is classified as either:

Normal

Anomaly

Dashboard Results

The dashboard provides visual analysis including:

- KPI Summary
- Throughput Trend
- Latency Trend
- SINR Distribution

- RSRP vs Throughput Scatter Plot
- Anomaly Distribution
- Downloadable Results

Limitations

This project has the following limitations:

- The dataset is relatively small
- Isolation Forest is an unsupervised model and does not provide anomaly categories
- Traditional evaluation metrics require labelled data
- The project uses offline CSV files rather than real-time network streams
- The application is deployed locally and not on a production cloud environment

Future Improvements

Future enhancements include:

- Real time KPI streaming
- Cloud deployment
- Automatic alert notifications
- Root Cause Analysis
- Database integration
- User authentication
- Comparison with One-Class SVM and Autoencoders

Conclusion

The Telecom KPI Anomaly Detection Dashboard successfully demonstrates the use of the Isolation Forest algorithm for identifying anomalous telecom KPI records. The combination of preprocessing, feature engineering, feature scaling, and machine learning enables efficient anomaly detection without requiring labelled training data. The interactive Streamlit dashboard further improves usability by providing engineers with real-time visual insights into network performance.